Unit 7 Guided Notes — Data Analysis & Statistics

Name _____ Date _____ Period _____

Scatterplots, correlation, the line of best fit, and two-way tables.

🎯 By the end of this unit, I can…

- Describe association in a scatter plot
- Draw and use a line of best fit to make predictions
- Read and build two-way tables and relative frequencies

Words to know

- **Bivariate data** — two numbers for each person or thing.
- **Scatter plot** — a graph that shows each pair of numbers as one dot.
- **Association** — two things change together in a pattern.
- **Outlier** — a dot far away from the rest.
- **Line of best fit** — one straight line through the middle of the dots.
- **Two-way table** — a table that counts people two ways at once.
- **Relative frequency** — a count divided by a total (a fraction or percent).
- **Lurking variable** — a hidden third thing that makes two things change together.

Standards: NC.8.SP.1 · NC.8.SP.2 · NC.8.SP.3 · NC.8.SP.4

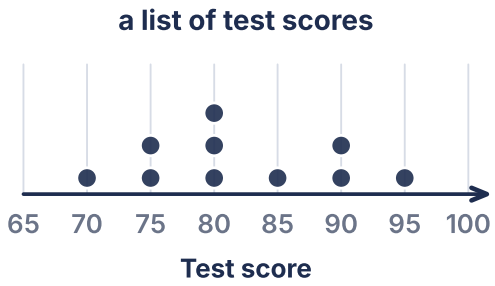
Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

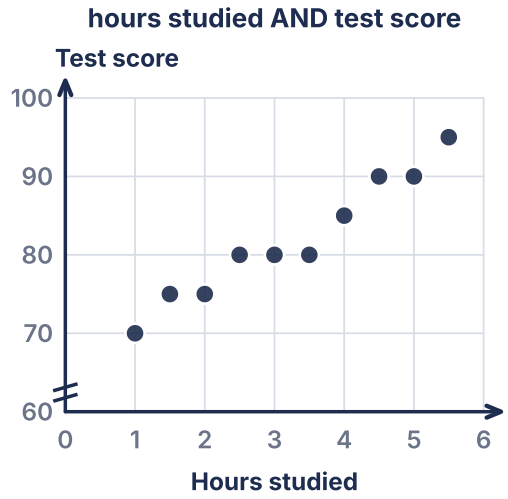
1 Scatterplots & Bivariate Data

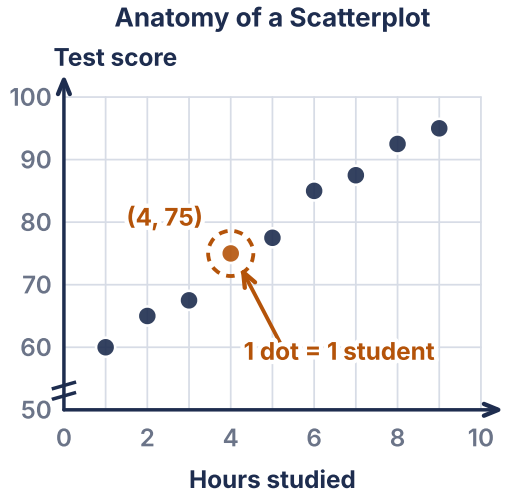
📌 **NC.8.SP.1:** Construct and interpret scatter plots for bivariate measurement data.
I can make a scatter plot and tell what it shows.

Bivariate data has TWO values for each item — like a person's height AND weight, or hours studied AND test score. We use a scatterplot to look for patterns between the two variables.

Term	Meaning	Example
Univariate data	_____	a list of test scores
Bivariate data	_____	hours studied AND test score





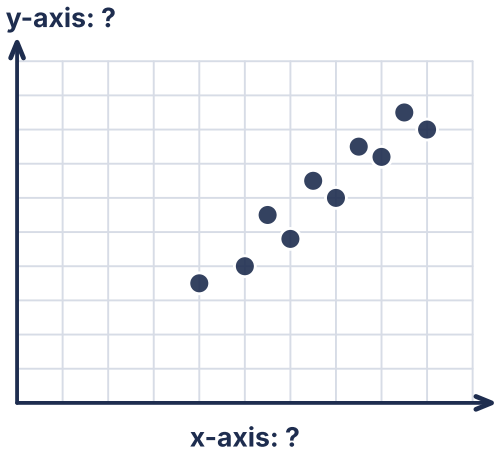


A scatterplot has: an x-axis for the _____ variable, a y-axis for the _____ variable. Each dot represents _____ data point. Both axes need a _____ and a _____.

📊 **The two variables together form an ordered pair (x, y) that we plot as a single dot.** Each dot is one person, one observation.

Try It

Suppose you collect data on 10 students: how many hours of sleep they got and their test score the next day.



Which data goes across? Which goes up? Count the dots.

- Independent variable (x-axis)? _____
- Dependent variable (y-axis)? _____
- Total number of dots on the scatterplot: _____

💡 **Why scatterplots?** Bar graphs count things in groups. Scatter plots show if two numbers change together. If you're asking "does X affect Y?", you want a scatterplot.

✏ Practice: Scatterplots

1. Score 3 Which data set is **bivariate**?

Heights

Student	Height (cm)
Ana	152
Ben	160
Cal	148

Sleep and test score

Student	Sleep (h)	Score
Ana	8	85
Ben	6	72
Cal	9	90

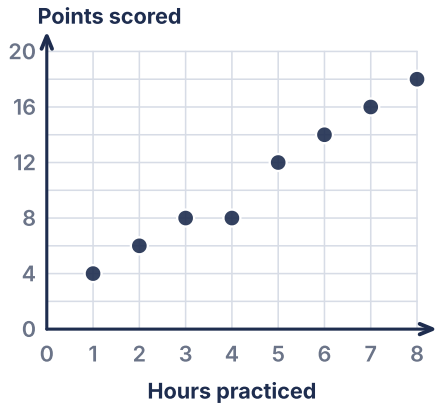
Favorite colors

Student	Color
Ana	blue
Ben	red
Cal	green

- ○ Heights of 20 students
- ○ Hours of sleep AND test score for 20 students
- ○ Favorite colors of 20 students

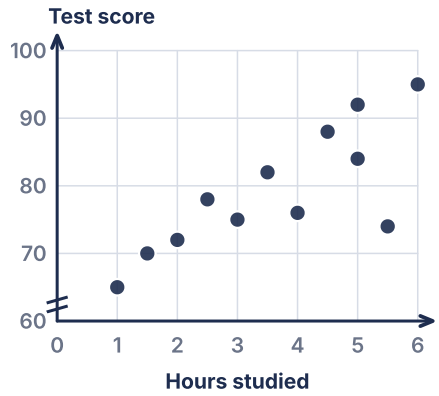
2. Score 4 The scatterplot shows hours of practice and points scored for 8 players. How many points did the player who practiced **5 hours** score?

Answer: _____



3. Score 5 How many students studied **more than 4 hours** AND scored **above 80**?

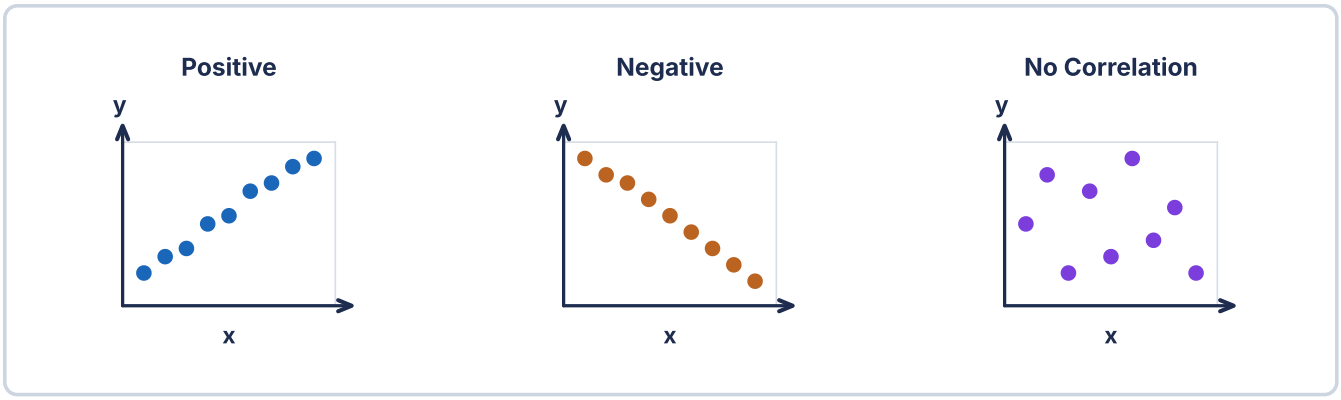
Answer: _____



2 Correlation

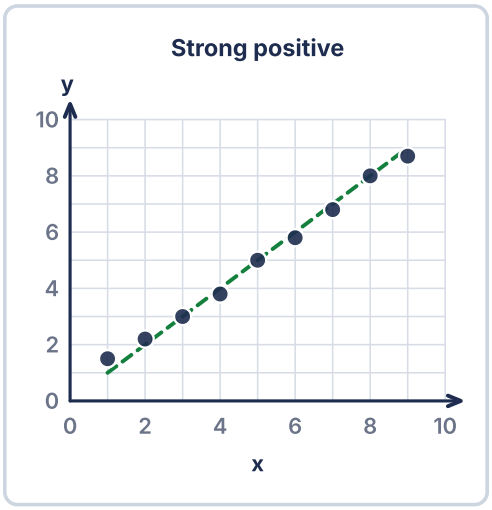
📌 **NC.8.SP.1:** Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.
I can tell if the dots go up, go down, or show no pattern, and spot outliers.

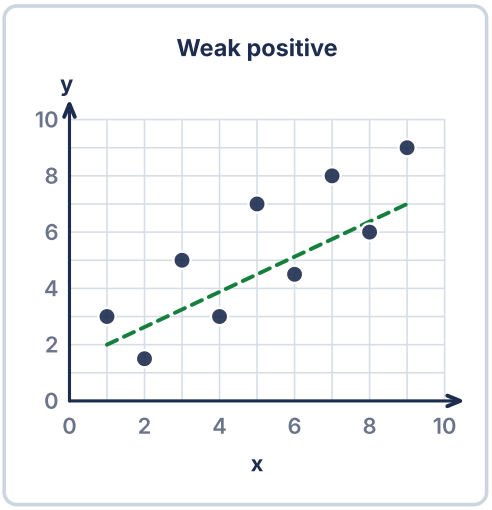
Correlation describes how two variables tend to change together — **positive**, **negative**, or **none**. Plus two strengths: **strong** or **weak**.



Positive correlation: as x increases, y also _____. Negative correlation: as x increases, y _____. No correlation: the dots are scattered with _____ clear pattern.

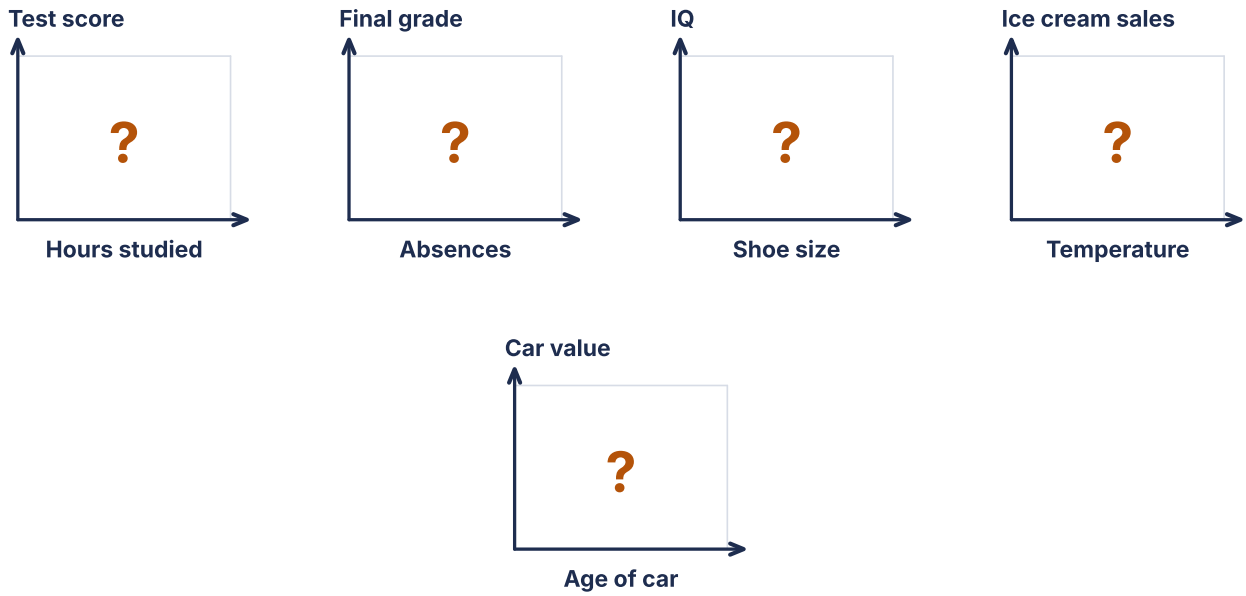
Strength of Correlation





Strong = dots tightly hug the trend line. Weak = dots are spread out around the trend.

Try It — Predict the correlation type

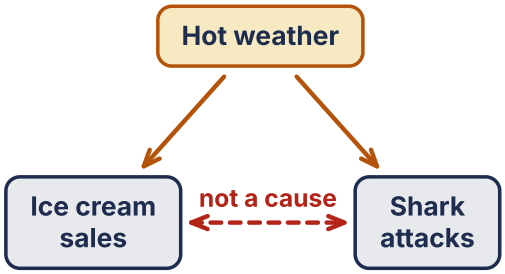


Picture the dots on each graph before you answer.

- Hours studied vs. test score → _____
- Number of absences vs. final grade → _____
- Shoe size vs. IQ → _____
- Outdoor temperature vs. ice cream sales → _____
- Age of a car vs. value of the car → _____

🎯 **Outliers and clusters:** An *outlier* is a dot that's far from the rest. A *cluster* is a group of dots tightly grouped together. Both are worth pointing out when you describe a scatterplot.

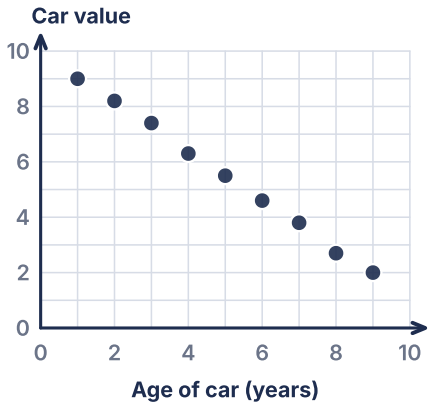
⚠️ **CORRELATION ≠ CAUSATION:** Just because two things are correlated doesn't mean one causes the other! Ice cream sales and shark attacks are both higher in summer — but ice cream doesn't cause sharks. Both are caused by hot weather.



✏ Practice: Correlation

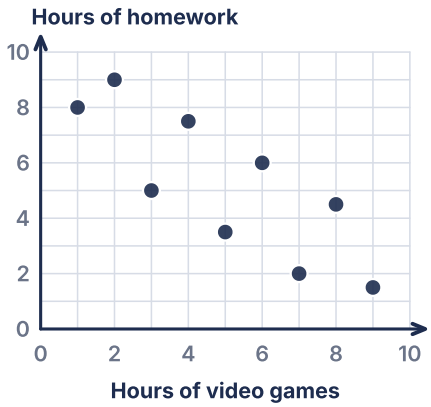
1. Score 3 What type of correlation does this scatterplot show?

- ○ Positive
- ○ Negative
- ○ None



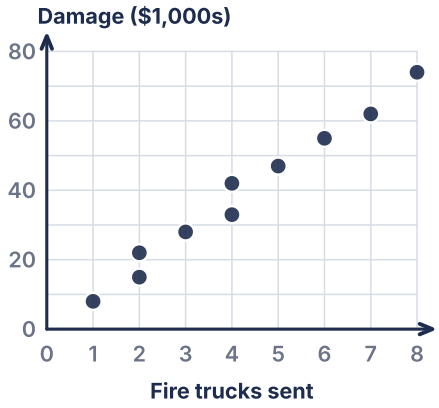
2. Score 4 Describe the correlation. Pick the direction AND the strength.

- ○ Strong positive
- ○ Weak positive
- ○ Strong negative
- ○ Weak negative



3. Score 5 Fires where more fire trucks show up have more damage. Does sending more trucks **cause** more damage?

- ○ Yes — more trucks cause more damage
- ○ No — big fires bring more trucks AND cause more damage
- ○ No — there is no correlation at all



3 Line of Best Fit

📌 **NC.8.SP.2 / NC.8.SP.3:** Informally fit a straight line for scatter plots that suggest a linear association; use the equation to solve problems and make predictions.
I can draw a line that fits the dots and use it to predict.

When the dots suggest a linear pattern, we can draw a single straight line through the "middle" of the cloud of dots. This line represents the overall trend and helps us make predictions.

When Should You Draw One?

A line of best fit only makes sense when:

- The data shows a _____ pattern (positive or negative correlation).
- You should NOT draw one if the dots show _____ correlation or a curved/non-linear pattern.

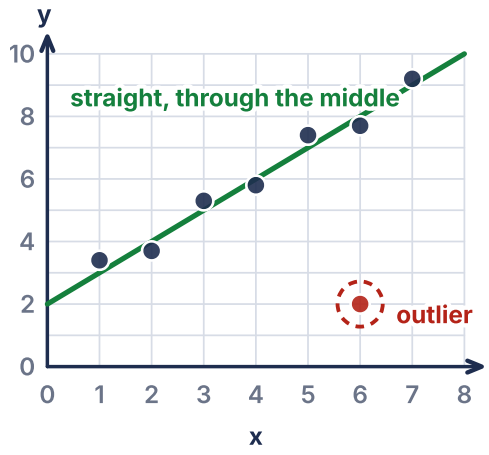


4 dots above, **4 dots below**, 1 right on the line.

Notice the line: passes through the "middle" of the dots · has roughly the same number of dots above and below it · follows the overall positive trend · doesn't have to touch any specific dot.

Drawing a Line of Best Fit (Informally)

1. Look at the overall direction of the dots (positive or negative).
2. Use a ruler — put the line so dots are roughly balanced above and below.
3. Don't connect the dots! It's a smooth straight line, not a zigzag.
4. Don't be tricked by **outliers** — ignore dots far from the rest.



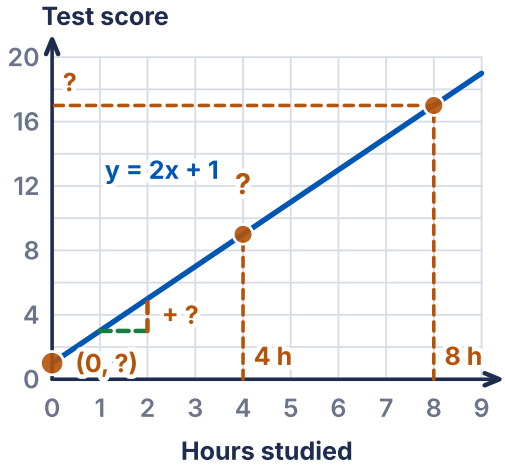
One ruler line. Dots balanced above and below. The red outlier is ignored.

Using the Line to Make Predictions

Once you draw a line of best fit, you can:

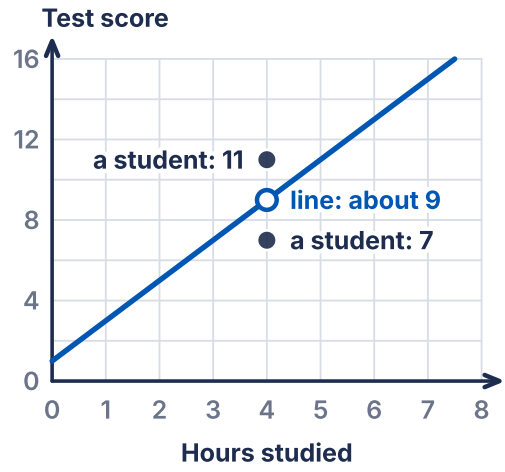
- Find its equation (**y = mx + b**) by reading off the slope and y-intercept.
- Plug in a new x-value to predict y.
- Read graphically: find x on the x-axis, go up to the line, then over to the y-axis.

Try It — y = 2x + 1 (x = hours studied, y = test score)



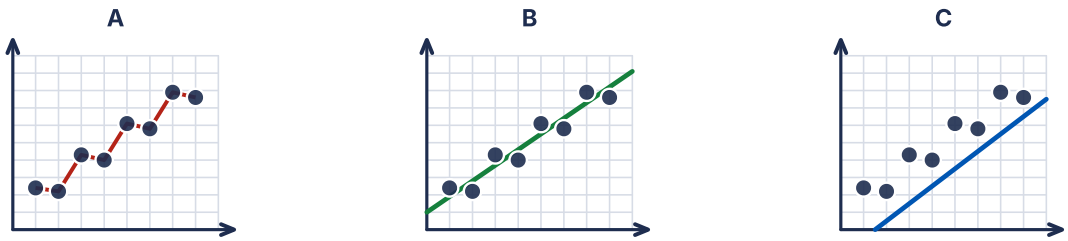
- The slope means: each extra hour of study adds about _____ points.
- The y-intercept means: with 0 hours of study, the predicted score is _____.
- If a student studies 4 hours, predicted score = 2(4) + 1 = _____.
- If a student studies 8 hours, predicted score = _____.

⚠️ **Predictions are approximate.** The line is a model, not exact. A student who studies 4 hours might get 7 or 11 — the line predicts about 9 on average.



✏ Practice: Line of Best Fit

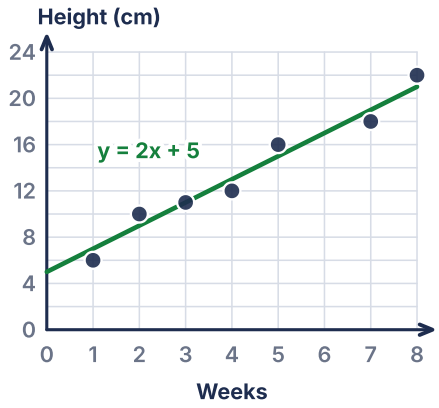
1. Score 3 Which graph shows a good line of best fit?



- ○ Graph A
- ○ Graph B
- ○ Graph C

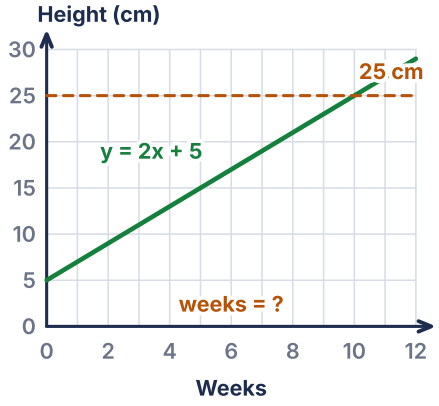
2. Score 4 The line of best fit for a plant is y = 2x + 5. x = weeks and y = height in cm. Predict the height at **6 weeks**.

Answer: _____



3. Score 5 Use the same line, y = 2x + 5. About how many **weeks** until the plant is **25 cm** tall?

Answer: _____



4 Two-Way Tables

📌 **NC.8.SP.4:** Understand patterns of association between two variables by displaying frequencies and relative frequencies in a two-way table.
I can read a two-way table and find percents in it.

A **two-way table** counts people two ways at once. Example: grade level AND whether they like pizza. Both things are groups (categories), not numbers.

When do we use them? Two-way tables help organize *categorical* bivariate data — when both variables are categories instead of numbers. Example: grade level and lunch preference, gender and favorite sport.

Reading a Two-Way Table — 120 students surveyed about pizza

	Like Pizza	Don't Like Pizza	Total
6th Grade	32	8	40
7th Grade	28	12	40
8th Grade	24	16	40
Total	**84**	**36**	**120**

	Like pizza	Do not like	Total
6th Grade	32	8	**40**
7th Grade	28	12	**40**
8th Grade	24	16	**40**
Total	**84**	**36**	**120**

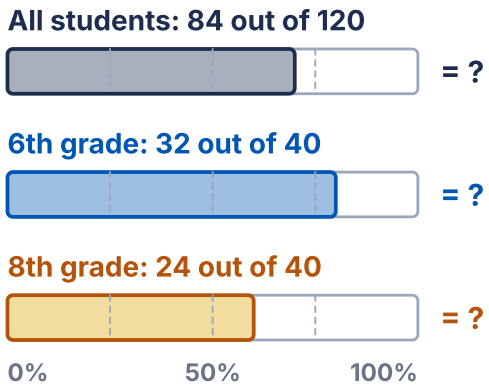
A row and a column cross at one cell: 6th grade AND likes pizza. Totals are on the edges.

- Total students surveyed: _____
- Number of 6th graders who like pizza: _____
- Total 7th graders surveyed: _____

- Total students who don't like pizza: _____

Relative Frequency

Relative frequency = the count divided by a total. Same idea as in Unit 6 frequency tables, but now we have multiple totals to choose from.



- Fraction of all 120 students who like pizza: 84 ÷ 120 = _____ or _____
- Fraction of 6th graders who like pizza: 32 ÷ 40 = _____ or _____
- Fraction of 8th graders who like pizza: 24 ÷ 40 = _____ or _____

💡 **Looking for an association:** 80% of 6th graders like pizza but only 60% of 8th graders do. There's an **association** between grade level and pizza preference — they're related!

Try It — 100 students surveyed about pets

	Has pet	No pet	Total
Boys	20	30	50
Girls	35	15	50
Total	**55**	**45**	**100**

	Has pet	No pet	Total
Boys	20	30	50
Girls	35	15	50
Total	55	45	100

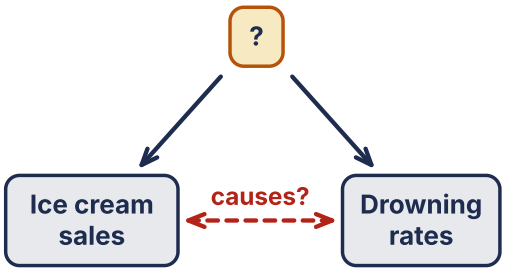
Boys: $\frac{20}{50} = ?$

Girls: $\frac{35}{50} = ?$

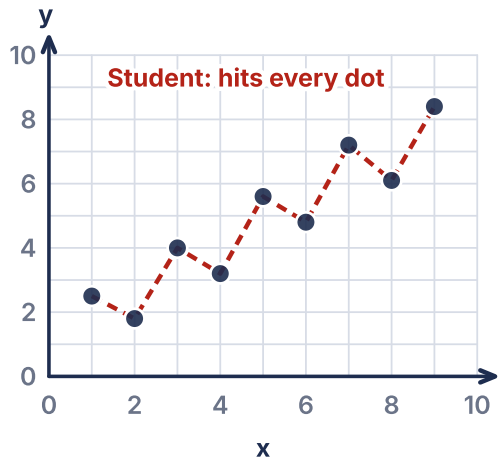
- Fraction of boys who have a pet: 20 ÷ 50 = _____ or _____
- Fraction of girls who have a pet: 35 ÷ 50 = _____ or _____
- Is there an association between gender and pet ownership? _____

Error Analysis

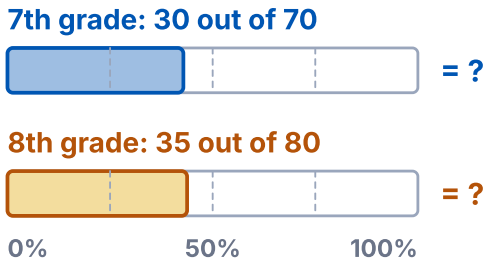
SPOT THE ERROR **"Ice cream sales and drowning rates have a positive correlation, so eating ice cream causes drowning."**



SPOT THE ERROR "The line of best fit must pass through every data point."



SPOT THE ERROR "More 8th graders like basketball (35) than 7th graders (30), so 8th graders are more into basketball." (8th grade has 80 students total vs 70 in 7th grade!)



✏ Practice: Two-Way Tables

1. Score 3 80 students were asked if they play a sport. How many **8th graders play a sport**?

	Plays a sport	No sport	Total
7th Grade	18	22	40
8th Grade	27	13	40
Total	**45**	**35**	**80**

Answer: _____

2. Score 4 Same table. What **percent of 8th graders** play a sport? Write it as a percent.

	Plays a sport	No sport	Total
7th Grade	18	22	40
8th Grade	27	13	40
Total	**45**	**35**	**80**

Answer: _____

3. Score 5 Same table. Is there an **association** between grade and playing a sport?

	Plays a sport	No sport	Total
7th Grade	18	22	40
8th Grade	27	13	40
Total	**45**	**35**	**80**

- ○ Yes — 67.5% of 8th graders play, but only 45% of 7th graders do
- ○ No — both grades have 40 students
- ○ No — the two grades have about the same percent

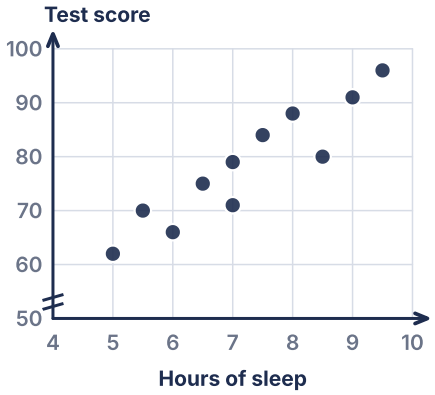
✏ Practice — All Sections

Mixed review of scatterplots, correlation, the line of best fit, and two-way tables. Look back at the notes if you get stuck.

✏ Comprehensive Practice All sections

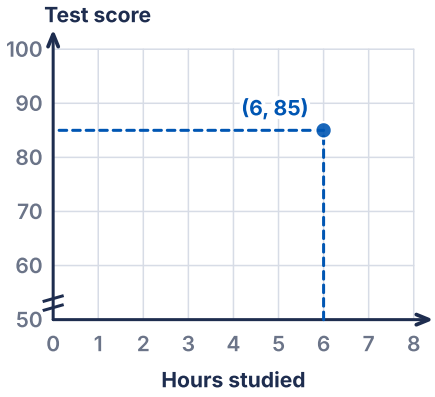
1. Score 3 Each dot is one student. How many students are shown?

Answer: _____



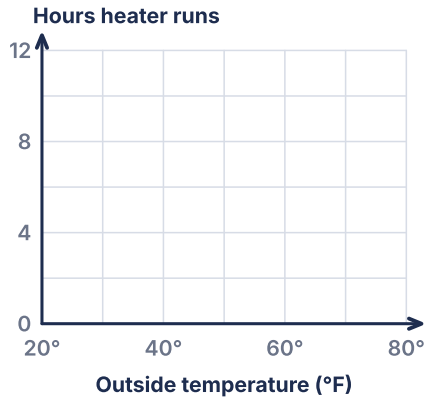
2. Score 3 On a scatterplot, x = hours studied and y = test score. What does the dot **(6, 85)** mean?

- ○ A student studied 85 hours and scored 6
- ○ A student studied 6 hours and scored 85
- ○ 6 students scored 85



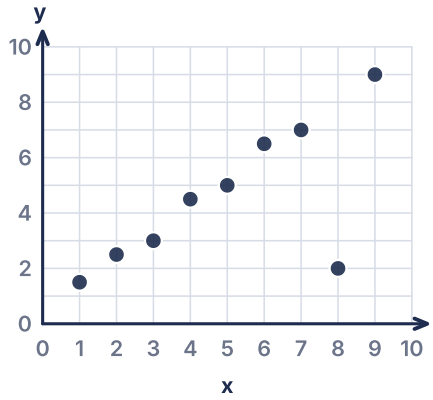
3. Score 3 Outside temperature vs. hours the heater runs. What type of correlation would you expect? Picture the dots on these axes.

- ○ Positive
- ○ Negative
- ○ None



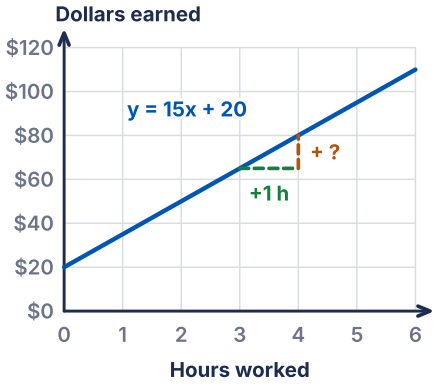
4. Score 4 One dot is an **outlier**. Write it as an ordered pair (x, y).

Answer: _____



5. Score 4 A line of best fit for Mia's pay is y = 15x + 20. x = hours worked and y = dollars earned. What does the **15** mean?

- ○ She earns $15 more for each extra hour
- ○ She starts with $15
- ○ She works 15 hours



6. Score 4 Find the missing number in the table.

	Has a phone	No phone	Total
6th Grade	12	?	30
7th Grade	21	9	30
Total	**33**	**27**	**60**

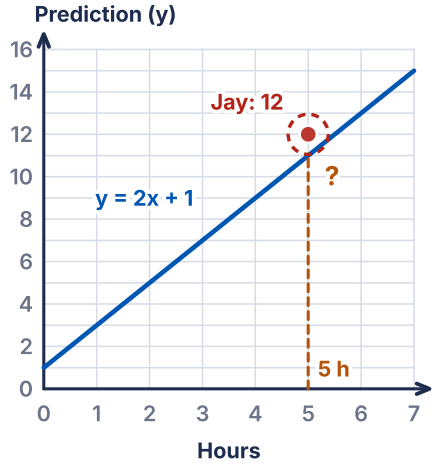
Answer: _____

7. Score 4 What **percent of girls** prefer cats? Write it as a percent.

	Dogs	Cats	Total
Boys	24	16	40
Girls	15	35	50
Total	**39**	**51**	**90**

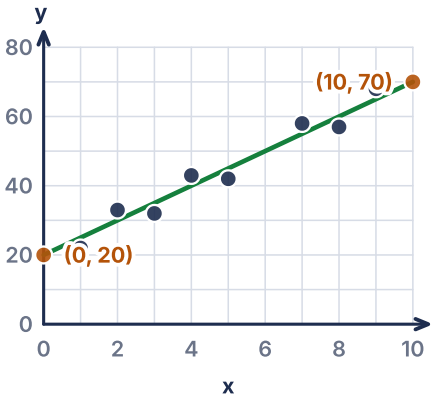
Answer: _____

8. Score 5 Jay used y = 2x + 1 for 5 hours and wrote 2 × 5 + 1 = 2 × 6 = 12. What is the correct prediction?



Answer: _____

9. Score 5 The line of best fit goes through (0, 20) and (10, 70). Predict y when **x = 6**.



Answer: _____

10. Score 5 Same table. Boys are more likely to prefer dogs. By how many **percentage points**?

	Dogs	Cats	Total
Boys	24	16	40
Girls	15	35	50
Total	**39**	**51**	**90**

Answer: _____